ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MULTIMEDIA BASICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 2. Section A: objective questions (30) · Section B: multimedia elements, images and colour, audio and video, design principles, hardware, copyright and the multimedia industry (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Multimedia combines …

A only text and images.
B two or more media such as text, images, audio, video and animation.
C only sound and video.
D only printed material.

1.1.2 An image made of tiny coloured squares (pixels) is a …

A vector image.
B font.
C raster (bitmap) image.
D codec.

1.1.3 Which file format supports transparency and lossless compression and is often used for logos on websites?

A PNG
B JPEG
C MP3
D BMP

1.1.4 The colour model used for screens is …

A CMYK.
B RGB.
C Pantone.
D greyscale.

1.1.5 Multimedia that lets the user control what happens (e.g. clicking buttons) is …

A linear.
B analogue.
C static.
D interactive.

1.1.6 The number of frames shown per second in a video is the …

A frame rate.
B resolution.
C bit depth.
D sample rate.

1.1.7 MP3 is a file format for …

A video.
B images.
C audio.
D animation.

1.1.8 A rough sequence of sketches that plans the scenes of a video or animation is a …

A template.
B watermark.
C storyboard.
D timeline.

1.1.9 Which device converts sound into an electrical signal?

A Microphone
B Speaker
C Projector
D Scanner

1.1.10 Lossy compression …

A never changes the quality.
B makes files larger.
C is used only for text.
D permanently removes some data to make files smaller.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Vector images lose quality when they are enlarged.

1.2.2 A higher resolution image contains more pixels.

1.2.3 CMYK is the colour model used for printing.

1.2.4 Sans-serif fonts such as Arial are generally easier to read on screen.

1.2.5 You may use any picture you find on the Internet in a commercial advert without permission.

1.2.6 A GIF file can contain a simple animation.

1.2.7 Stereo sound uses one channel only.

1.2.8 Contrast in design helps important elements to stand out.

1.2.9 A codec compresses and decompresses audio or video.

1.2.10 Alternative text (alt text) on images helps people who use screen readers.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	The smallest element of a digital image	A	sample rate
1.3.2	Number of bits used for the colour of each pixel	B	white space
1.3.3	Legal right of a creator over his or her work	C	Creative Commons
1.3.4	Licence that lets others use work under certain conditions	D	colour depth
1.3.5	Text or logo placed over an image to show ownership	E	pixel
1.3.6	Link in a multimedia product that jumps to another page	F	hyperlink
1.3.7	Area of a page left open to make the design less cluttered	G	copyright
1.3.8	Moving graphics created from a series of images	H	watermark
1.3.9	Number of times per second sound is measured when it is recorded	I	animation
1.3.10	The person who decides what a client needs and designs the product's look	J	multimedia designer

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MULTIMEDIA ELEMENTS AND APPLICATIONS

2.1 Name the FIVE basic elements of multimedia. (5)

2.2 Name FOUR areas where multimedia is used and give an example of each. (8)

2.3 Explain the difference between linear and non-linear (interactive) multimedia. Give an example of each. (4)

2.4 Give THREE advantages of using multimedia in education. (3)

[20]

QUESTION 3: IMAGES AND COLOUR

3.1 Compare raster and vector images under the following headings: (a) how the image is made; (b) what happens when it is enlarged; (c) a suitable use. (6)

3.2 An image is 1 200 pixels wide and 800 pixels high and uses 24-bit colour. Calculate its uncompressed file size in megabytes (MB). (1 MB = 1 024 × 1 024 bytes) (4)

3.3 Recommend a suitable file format for EACH of the following and give a reason: (a) a holiday photograph for a website; (b) a company logo with a transparent background. (4)

3.4 Explain why a designer would use CMYK and not RGB for a poster that will be printed. (2)

3.5 What is meant by *resolution* of a printed image, and why does a print need a higher resolution (e.g. 300 dpi) than a screen? (4)

[20]

QUESTION 4: AUDIO, VIDEO AND HARDWARE

4.1 Explain why video files are usually compressed. (2)

4.2 Name THREE hardware devices needed to record and edit a short promotional video. (3)

4.3 Give TWO factors that influence the quality of a digital sound recording. (2)

4.4 Explain the purpose of a storyboard in a video project. (3)

[10]

QUESTION 5: DESIGN, COPYRIGHT AND THE INDUSTRY

5.1 A local spaza shop asks you to design a one-page digital flyer. Name and explain FOUR design principles you will use. (8)

5.2 Explain what copyright is, and give TWO ways a designer can use images legally. (4)

5.3 Give FOUR steps in planning a multimedia project, in order. (4)

5.4 Name TWO careers in the multimedia industry. (2)

5.5 Give TWO ways to make a multimedia product accessible to people with disabilities. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER